PURPOSE: This procedure establishes the controlled shutdown, physical preventive maintenance, and systematic restart of all Power Distribution Units, UPS systems, MDP panels, WAB panels, and the LVMSB-3 switchboard serving Data Center 1 at the TIM Data Center 1 facility, while maintaining redundant power to all client equipment throughout the activity.

SCOPE: This MOP covers all electrical distribution and UPS equipment within the TIM Data Center 1 Main Building and Annex Building A-2F, specifically:

- PDU 13, PDU 14, PDU 16, PDU 17 (Data Center Floor, Main Building)
- PDU 13A, PDU 14A, PDU 16A (EE Room 3 / Telco, Main Building)
- MDP-UPS-O/P-3 Output Main Distribution Panel (EE Room 3, Main Building)
- MDP-UPS-I/P-3 Input Main Distribution Panel (UPS Room 3, Annex Building A-2F)
- UPS-3A, UPS-3B, UPS-3C (UPS Room 3, Annex Building A-2F)
- Battery Banks — System 3 (Battery Room 3, Annex Building A-2F)
- WAB 3 Panel (UPS Room 3, Annex Building A-2F)
- LVMSB-3 Low Voltage Main Switchboard (LVMSB Room 3, Annex Building A-2F)
- TIE-BREAKER / Bypass Breaker located at LVMSB-2 Panel (LVMSB Room 2, Annex Building A-2F)
- Genset 1 (Primary backup generation) and Genset 5 (Fallback via MISS system)

This MOP does not cover equipment tagged Red or Orange (will not be maintained) per pre-activity tagging conducted by the Data Center Operations (DCO) team.

PREREQUISITES: All of the following conditions must be confirmed true before any step of this procedure begins:

1. Client advisory emails have been issued at the 3-month, 2-month, and 1-month intervals prior to this activity date of June 05, 2026. All client confirmations have been collected by TIM Account Managers.
2. All client special requests have been received and reviewed no later than 3 days prior to this activity. No client requests are to be accepted on the day of the activity.
3. The EPM vendor (Exquis) has completed the pre-thermal scan of all in-scope equipment within the 2–4 week window prior to June 05, 2026. Thermal scan results have been reviewed and any critical findings have been escalated and resolved or formally accepted.
4. The DCO team has completed an electrical load analysis within the 1–4 week window prior to June 05, 2026. Load analysis results confirm redundant PDU pairs (PDU 12, PDU 11, PDU 3, PDU 2) have sufficient capacity to absorb full transferred load.
5. The DCO has compiled and distributed the rectification items list within the 2–4 week window prior to this activity.
6. The DCO has held the vendor preparation meeting with Exquis (and Blackstone where applicable) no later than 1 week prior to June 05, 2026.
7. Equipment tagging has been completed by the DCO no later than 3 days before this activity:
   - Yellow tags = equipment that will be maintained (in scope)
   - Red / Orange tags = equipment that will not be maintained (out of scope — do not work on these)
8. The equipment list for all client racks has been verified and updated within 1 month of this activity, confirming quantity, power source configuration (dual-fed or single-fed), and brand/model for each

rack served by in-scope PDUs.

9. Genset 1 has been tested and confirmed operational. Genset 5 has been confirmed available as fallback via the MISS system.
10. All required personnel are on-site, badged, and confirmed in their assigned team roles prior to the toolbox meeting.
11. The DCO has conducted the toolbox meeting no later than 1 hour before the start of this procedure. All team leads have signed the toolbox meeting attendance log.
12. An advisory has been sent to TIM Teams channel and directly to all affected clients confirming the activity is about to begin.
13. All required PPE (see below) has been inspected and is on-site and available for all personnel.
14. All required tools, torque wrenches, test equipment, and replacement dead front panels are staged and confirmed on-site.
15. A communication channel (radio or equivalent) is active and confirmed working between Team 1 (Control), Team 2, Team 3, Team 4, Team 5, and Team 6 before the first step is executed.

REQUIRED PPE: All personnel working within electrical rooms, the UPS room, battery room, LVMSB room, or on or near any energized or recently de-energized equipment must wear the following at minimum:

- Arc-flash rated personal protective equipment (PPE) at the arc flash incident energy level calculated for each work location per the facility arc flash hazard analysis
- Arc-flash rated face shield or arc-rated balaclava with arc-rated face shield (rated for applicable incident energy category)
- Arc-rated gloves rated for the voltage class of equipment being worked on (minimum Class 00 for 400V equipment; verify per arc flash study)
- Arc-rated long-sleeve shirt and arc-rated pants or arc-rated coverall (minimum Arc Rating matching incident energy)
- Hard hat (non-conductive, Class E electrical rated)
- Safety glasses worn under face shield at all times
- Leather over-boots or dielectric safety boots (non-conductive soles) rated for the work environment
- High-visibility safety vest when working in areas with vehicle or equipment traffic
- Hearing protection where required by facility noise level assessments

Note: PPE requirements for battery room work additionally require:**

- Chemical splash goggles (in addition to safety glasses) due to electrolyte hazard
- Chemical-resistant gloves over arc-rated gloves when handling battery connections
- Chemical-resistant apron

SAFETY WARNINGS:

⚠ WARNING: THIS PROCEDURE INVOLVES WORK ON ENERGIZED AND RECENTLY DE-ENERGIZED ELECTRICAL EQUIPMENT OPERATING AT 400V LINE-TO-LINE AND ABOVE. CONTACT WITH ENERGIZED CONDUCTORS AT THESE VOLTAGE LEVELS IS POTENTIALLY FATAL. ALL PERSONNEL MUST MAINTAIN THE REQUIRED APPROACH BOUNDARIES ESTABLISHED BY THE FACILITY ARC FLASH HAZARD ANALYSIS AT ALL TIMES.

⚠ WARNING: ARC FLASH HAZARD IS PRESENT AT ALL PANEL INTERIORS, SWITCHBOARD COMPARTMENTS, UPS INPUT/OUTPUT TERMINALS, AND MDP ENCLOSURES. AN ARC FLASH EVENT AT THESE EQUIPMENT RATINGS CAN CAUSE SEVERE BURNS, BLINDNESS, BLAST INJURY, AND DEATH. NO PANEL COVER OR DEAD FRONT SHALL BE REMOVED WITHOUT CONFIRMING FULL PPE IS WORN BY ALL PERSONNEL IN THE IMMEDIATE AREA.

⚠ WARNING: THE UPS SYSTEMS (UPS-3A, UPS-3B, UPS-3C) REMAIN ENERGIZED ON BATTERY AND IN STS (STATIC TRANSFER SWITCH) MODE AFTER THE INVERTER IS TURNED OFF AND THE INPUT BREAKER IS OPENED. THE UPS OUTPUT IS STILL LIVE IN THIS STATE. DO NOT TREAT THE UPS AS DE-ENERGIZED UNTIL ALL BATTERY BANK BREAKERS IN BATTERY ROOM 3 HAVE BEEN CONFIRMED OPEN AND VOLTAGE HAS BEEN TESTED AND CONFIRMED ABSENT AT THE UPS OUTPUT TERMINALS.

⚠ WARNING: BATTERY BANKS IN BATTERY ROOM 3 CONTAIN HIGH-ENERGY DC VOLTAGE. BATTERY CIRCUITS CAN DELIVER MASSIVE SHORT-CIRCUIT CURRENT CAPABLE OF CAUSING FIRE, EXPLOSION, AND FATAL ELECTROCUTION. DO NOT WEAR METALLIC JEWELRY, RINGS, WATCHES, OR CONDUCTIVE ACCESSORIES IN THE BATTERY ROOM AT ANY TIME.

⚠ WARNING: BATTERY BANKS EMIT HYDROGEN GAS DURING NORMAL OPERATION. HYDROGEN IS EXPLOSIVE. CONFIRM BATTERY ROOM 3 VENTILATION IS OPERATING NORMALLY BEFORE ENTERING AND DURING ALL WORK IN THE BATTERY ROOM. DO NOT INTRODUCE ANY IGNITION SOURCE INTO THE BATTERY ROOM.

⚠ WARNING: SINGLE-FED CLIENT EQUIPMENT WILL LOSE POWER WHEN ITS PDU BRANCH CIRCUIT IS OPENED. CONFIRM THE CURRENT EQUIPMENT LIST FOR EACH CLIENT RACK BEFORE OPENING ANY BRANCH CIRCUIT. IF ANY SINGLE-FED EQUIPMENT IS IDENTIFIED THAT WAS NOT PREVIOUSLY DISCLOSED, STOP, REPORT TO TEAM 1 CONTROL, AND AWAIT DIRECTION BEFORE PROCEEDING.

⚠ WARNING: BRANCH CIRCUITS MUST BE OPENED ONE AT A TIME PER CLIENT RACK IN THE SEQUENCE DIRECTED BY TEAM 1 CONTROL. BATCH-TRIPPING MULTIPLE BRANCH CIRCUITS SIMULTANEOUSLY IS STRICTLY PROHIBITED AND MAY CAUSE UNCONTROLLED LOAD DROPS TO CLIENT EQUIPMENT.

⚠ WARNING: LVMSB-3 FEEDS ALL PDUs, UPS INPUTS, MDP PANELS, AND WAB PANELS IN SCOPE. OPENING THE TIE-BREAKER AT LVMSB-2 IS AN IRREVERSIBLE STEP DURING THIS PROCEDURE THAT REMOVES THE UPSTREAM FEED TO THE ENTIRE SYSTEM 3 DISTRIBUTION CHAIN. CONFIRM ALL DOWNSTREAM EQUIPMENT HAS BEEN PROPERLY ISOLATED AND ALL CLIENT LOADS HAVE BEEN VERIFIED AS TRANSFERRED TO REDUNDANT FEEDS BEFORE THIS STEP IS EXECUTED.

⚠ WARNING: DO NOT RE-ENERGIZE ANY PANEL, PDU, OR UPS SYSTEM WITHOUT EXPLICIT VERBAL AUTHORIZATION FROM TEAM 1 CONTROL. UNAUTHORIZED RE-ENERGIZATION OF EQUIPMENT DURING MAINTENANCE ACTIVITIES CAN CAUSE INJURY TO PERSONNEL WORKING INSIDE PANELS AND CAN DAMAGE CLIENT EQUIPMENT.

⚠ WARNING: IF AT ANY POINT GENSET 1 FAILS TO SUPPORT THE LOAD DURING THIS ACTIVITY, IMMEDIATELY NOTIFY TEAM 1 CONTROL. THE FALLBACK PATH IS GENSET 5 VIA THE MISS SYSTEM. DO NOT ATTEMPT ANY GENERATOR SWITCHING WITHOUT AUTHORIZATION FROM THE FACILITY ENGINEER.

⚠ WARNING: ALL LOCKOUT/TAGOUT (LOTO) PROCEDURES REQUIRED BY FACILITY SAFETY POLICY AND APPLICABLE REGULATORY REQUIREMENTS MUST BE APPLIED AT EVERY STEP WHERE EQUIPMENT IS DE-ENERGIZED FOR MAINTENANCE. DO NOT REMOVE ANY LOCKOUT OR TAGOUT DEVICE UNTIL MAINTENANCE IS COMPLETE AND THE TEAM LEAD RESPONSIBLE FOR THAT EQUIPMENT HAS GIVEN EXPLICIT AUTHORIZATION.

ESTIMATED DURATION: Total estimated activity duration: approximately 8–10 hours for the full sequence including all PDU shutdowns, maintenance, UPS system shutdown, panel maintenance, and full system restart and verification. Individual task durations are noted within the procedure steps. The toolbox meeting conducted 1 hour before the start adds additional time prior to clock start on the procedure itself.

Breakdown reference:

- PDU dead front replacement: 30 minutes each (PDU 13, PDU 16 confirmed; applied as standard for all in-scope PDUs)
- PDU branch circuit maintenance: 15 minutes per PDU (PDU 14, 16, 17 and others)
- PDU 13A / 14A / 16A maintenance: 45 minutes (EE Room 3)
- LVMSB-3 maintenance: 80 minutes
- MDP-UPS-O/P-3 dead front replacement: 30 minutes (parallel with LVMSB-3 maintenance)
- WAB-3 dead front replacement: 30 minutes (parallel with LVMSB-3 maintenance)
- MDP-UPS-O/P-3 panel maintenance: 35 minutes
- MDP-UPS-I/P-3 dead front replacement: 30 minutes (parallel with MDP-UPS-O/P-3 maintenance)
- WAB-3 panel maintenance: 35 minutes
- LVMSB-3 dead front replacement: 30 minutes (parallel with WAB-3 maintenance)
- MDP-UPS-I/P-3 panel maintenance: 35 minutes

REQUIRED PERSONNEL: Minimum staffing required for this procedure. All roles must be filled before the procedure begins. No step shall be executed without the assigned team in position and in radio communication with Team 1 Control.

- **Team 1 — Control Lead (1 person minimum):** Senior Data Center Operations engineer or qualified DCO Shift Lead. This person has sole authority to authorize each sequential phase of this procedure. Stationed at the Control point with visibility to the monitoring system and communication with all teams.
- **Team 2 — Switching Team (2 persons minimum):** Qualified electricians with authorization to operate distribution breakers and PDU branch circuits. Responsible for executing all switching operations at PDUs, MDP panels, WAB panels, LVMSB, and UPS systems. At minimum one Exquis vendor technician shall be part of or directly paired with this team during UPS and LVMSB operations.
- **Team 3 — Verification Team (2 persons minimum):** Qualified electricians or DCO engineers responsible for metering, voltage verification, and confirming load transfer on the monitoring system and at panel meters. Responsible for pre-maintenance safety voltage verification.
- **Team 4 — Floor Patrol Team A (2 persons minimum):** DCO staff qualified to visually inspect rack equipment and identify equipment that has gone offline. Covers assigned aisles during all PDU branch circuit switching operations.

- **Team 5 — Floor Patrol Team B (2 persons minimum):** Same qualifications as Team 4. Covers separate assigned aisles simultaneous to Team 4.
- **Team 6 — Maintenance Team (Exquis vendor lead + technicians as required):** Qualified electrical maintenance technicians from Exquis (primary EPM vendor). Responsible for all physical preventive maintenance tasks on PDUs, panels, UPS systems, and battery banks. Blackstone technicians shall assist with PDU 16 maintenance and verification activities as previously coordinated.
- **Client Services Group (CSG) Liaison (1 person):** TIM staff member available throughout the activity to receive reports from Teams 4 and 5 regarding any client equipment found offline and to communicate immediately with the affected client.

STEP BY STEP PROCEDURE:

PHASE 1 — FINAL PRE-ACTIVITY CONFIRMATION

1. Team 1 Control calls roll for all teams via radio. Confirm Team 2, Team 3, Team 4, Team 5, Team 6, and the CSG Liaison are all in position and radio communication is confirmed working in both directions with each team before proceeding.

2. Team 1 Control confirms with the DCO monitoring system operator that the real-time power monitoring dashboard is active and all in-scope PDU load readings, UPS system status, and generator status are visible and displaying normal values.

3. Team 1 Control confirms with Team 6 (Exquis lead) that all maintenance tools, torque wrenches, cleaning materials, thermal camera, digital multimeter, and replacement dead front panels for all in-scope equipment are staged and inventoried on-site.

4. Team 1 Control verbally confirms with Team 2 that all equipment tagged Red or Orange is identified and excluded from all switching operations. Team 2 lead verbally acknowledges the list of Red and Orange tagged equipment by name.

5. Team 1 Control sends the final activity start advisory message to the TIM Teams channel and to all affected client contacts confirming the EPM activity is now commencing at TIM Data Center 1.

> ✅ **VERIFY CHECKPOINT 1:** Confirm all six teams plus CSG Liaison have checked in by radio. Confirm monitoring dashboard is active. Confirm all tools and replacement parts are staged. Confirm Red/Orange exclusions are acknowledged. Confirm client advisory has been sent. Do not proceed until all five conditions are confirmed.

PHASE 2 — PDU 13 SHUTDOWN AND MAINTENANCE

6. Team 1 Control authorizes Team 2 to begin PDU 13 branch circuit shutdown sequence. Team 2 proceeds to PDU 13 on the Data Center floor, Main Building.

7. Team 3 confirms on the monitoring system that PDU 12 (redundant pair for PDU 13) is live, carrying its current load normally, and shows available capacity to accept transferred load from PDU 13 before Team 2 opens any branch circuit.

8. Team 2 opens PDU 13 Branch Circuit 1 (first circuit, client rack assigned). Open one circuit only. Do not open additional circuits until step 9 is complete.

9. Teams 4 and 5 simultaneously walk the assigned rack aisles and visually confirm that the client rack served by the branch circuit opened in Step 8 shows no equipment has gone offline. Team 4 lead and Team 5 lead each radio their confirmation to Team 1 Control.

10. If any equipment is found offline in Step 9: Team 4 or Team 5 immediately reports the rack location and equipment description to the CSG Liaison. The CSG Liaison contacts the affected client to confirm whether operations are impacted. Team 1 Control places a hold on further PDU 13 branch circuit openings until the CSG Liaison confirms the client's status and Team 1 Control explicitly authorizes resuming. If equipment is found offline and the client confirms operational impact, execute the ABORT procedure in the IF SOMETHING GOES WRONG section.

> ✅ **VERIFY CHECKPOINT 2:** Confirm PDU 12 load has increased to reflect transferred load from the circuit opened in Step 8. Confirm no equipment is offline in rack aisles served by PDU 13. Confirm no holds are active from Team 4, Team 5, or CSG. Authorize Team 2 to continue to next circuit only after all confirmations are received.

11. Team 2 continues opening PDU 13 branch circuits one at a time, following the same individual-circuit-open → floor patrol confirm sequence (Steps 8 through 10 repeated) for each of the remaining 29 branch circuits of PDU 13. Team 1 Control provides explicit authorization before each circuit opening. Teams 4 and 5 conduct a visual walk confirmation after every circuit opening.

12. After all 30 branch circuits of PDU 13 are confirmed open and all floor patrol checks are complete, Team 2 opens the PDU 13 main breaker. Team 2 lead radios Team 1 Control to confirm PDU 13 main breaker is open.

13. Team 1 Control, upon receiving confirmation from Team 2 that PDU 13 main breaker is open, proceeds to EE Room 3 (or designates a qualified Team 2 member already at EE Room 3) and opens the PDU 13 feeder breaker at MDP-UPS-O/P-3. Team 1 Control radios Team 6 to confirm PDU 13 feeder is open at MDP-UPS-O/P-3.

14. Team 3 verifies with a calibrated digital multimeter at the PDU 13 main terminals that voltage is absent (0V confirmed line-to-line and line-to-neutral) before Team 6 is authorized to open the PDU 13 enclosure. Team 3 lead radios confirmation to Team 1 Control.

15. Team 1 Control applies LOTO at PDU 13 main breaker and at the PDU 13 feeder breaker at MDP-UPS-O/P-3 per facility LOTO policy. Confirm LOTO tags and locks are applied and documented before authorizing Team 6 to begin maintenance.

> ✅ **VERIFY CHECKPOINT 3:** Confirm PDU 13 main breaker is open. Confirm PDU 13 feeder breaker at MDP-UPS-O/P-3 is open. Confirm 0V verified by Team 3 at PDU 13 terminals. Confirm LOTO applied at both isolation points. Authorize Team 6 to begin physical maintenance on PDU 13 only after all four conditions are confirmed.

16. Team 6 (Exquis, with Blackstone support for PDU 16 as applicable to this phase) performs physical preventive maintenance on PDU 13. Tasks include: dead front panel replacement (estimated 30 minutes), internal inspection, cleaning, connection torque verification, and thermal imaging of internal connections. Team 6 lead records all findings on the maintenance log.

17. Team 6 lead notifies Team 1 Control upon completion of all PDU 13 maintenance tasks. Team 6 confirms dead front is reinstalled and secured, no tools or foreign materials are inside the enclosure, and the enclosure door is closed.

18. Team 1 Control removes LOTO from PDU 13 main breaker and PDU 13 feeder breaker at MDP-UPS-O/P-3 per facility LOTO policy. Document LOTO removal in the activity log.

19. Team 3 confirms monitoring system shows PDU 13 feeder breaker status ready to close and no fault conditions indicated at PDU 13.

20. Team 1 Control closes the PDU 13 feeder breaker at MDP-UPS-O/P-3, then authorizes Team 2 to close the PDU 13 main breaker. Team 2 closes the PDU 13 main breaker and radios confirmation to Team 1 Control.

> ✅ **VERIFY CHECKPOINT 4:** Confirm PDU 13 feeder closed at MDP-UPS-O/P-3. Confirm PDU 13 main breaker closed. Confirm monitoring system shows PDU 13 energized and voltage reading at normal levels. Confirm no alarms at PDU 13. Do not restore branch circuits until all four conditions are confirmed.

21. Team 2 restores PDU 13 branch circuits one at a time per client rack. After closing each branch circuit, Teams 4 and 5 confirm the associated rack equipment is online and operating normally before the next circuit is closed. Team 1 Control authorizes each individual branch circuit closure.

22. After all 30 PDU 13 branch circuits are restored and all client rack equipment in the PDU 13 served aisles is confirmed online by Teams 4 and 5, Team 1 Control declares PDU 13 maintenance and restoration complete. Record completion time in the activity log.

PHASE 3 — PDU 14 SHUTDOWN AND MAINTENANCE

23. Team 1 Control authorizes Team 2 to begin PDU 14 branch circuit shutdown sequence. Team 2 proceeds to PDU 14 on the Data Center floor.

24. Team 3 confirms on the monitoring system that PDU 11 (redundant pair for PDU 14) is live and has confirmed available capacity to accept full transferred load from PDU 14 before any branch circuit is opened.

25. Team 2 opens PDU 14 branch circuits one at a time per client rack (60 total circuits). The same individual-circuit-open → Teams 4 and 5 floor patrol confirm sequence used in Phase 2 (Steps 8 through 10 procedure) applies to every circuit. Team 1 Control provides explicit authorization before each circuit is opened. Any equipment found offline is reported immediately to CSG Liaison per the process defined in Step 10.

> ✅ **VERIFY CHECKPOINT 5:** After all 60 PDU 14 branch circuits are confirmed open: confirm PDU 11 load has increased appropriately, confirm no equipment is offline in PDU 14-served aisles, confirm no holds are active. Authorize Team 2 to open PDU 14 main breaker only after all conditions are confirmed.

26. Team 2 opens the PDU 14 main breaker and radios confirmation to Team 1 Control.

27. Team 1 Control opens (or directs Team 2 to open) the PDU 14 feeder breaker at MDP-UPS-O/P-3 in EE Room 3. Confirmation is radioed to Team 6.

28. Team 3 verifies 0V at PDU 14 main terminals with a calibrated digital multimeter. Team 3 radios confirmation to Team 1 Control.

29. Team 1 Control applies LOTO at PDU 14 main breaker and PDU 14 feeder breaker at MDP-UPS-O/P-3. Document LOTO application in the activity log.

30. Team 6 performs physical preventive maintenance on PDU 14. Tasks include: dead front panel replacement (estimated 30 minutes), branch circuit maintenance (estimated 15 minutes), internal inspection, cleaning, connection torque verification, and thermal imaging. Team 6 lead records all findings.

> ✅ **VERIFY CHECKPOINT 6:** Confirm PDU 14 maintenance is complete per Team 6 lead confirmation. Confirm dead front is reinstalled and secured. Confirm no tools or foreign material inside enclosure. Confirm LOTO is ready for removal per Team 1 Control authorization. Do not proceed to PDU 14 restoration until all conditions are confirmed.

31. Team 1 Control removes LOTO from PDU 14 main breaker and PDU 14 feeder breaker at MDP-UPS-O/P-3. Document removal.

32. Team 1 Control closes PDU 14 feeder breaker at MDP-UPS-O/P-3. Team 2 closes PDU 14 main breaker on authorization from Team 1 Control.

33. Team 3 confirms PDU 14 is energized on the monitoring system, voltage is at normal levels, and no alarms are present at PDU 14.

34. Team 2 restores PDU 14 branch circuits one at a time per client rack (60 circuits). Teams 4 and 5 confirm each rack is operational after each circuit is restored. Team 1 Control authorizes each individual closure.

35. After all 60 PDU 14 branch circuits are restored and all client rack equipment in PDU 14-served aisles is confirmed online, Team 1 Control declares PDU 14 maintenance and restoration complete. Record completion time.

> ✅ **VERIFY CHECKPOINT 7:** Confirm all 60 PDU 14 branch circuits restored. Confirm Teams 4 and 5 have cleared all PDU 14-served aisles with no offline equipment found. Confirm monitoring system shows PDU 14 at normal load and no alarms. Confirm PDU 14 restoration is complete in activity log before proceeding to Phase 4.

PHASE 4 — PDU 16 SHUTDOWN AND MAINTENANCE

36. Team 1 Control authorizes Team 2 to begin PDU 16 branch circuit shutdown sequence. Team 2 proceeds to PDU 16 on the Data Center floor.

37. Team 3 confirms PDU 3 (redundant pair for PDU 16) is live and has confirmed available capacity before any PDU 16 branch circuit is opened.

38. Team 2 opens PDU 16 branch circuits one at a time per client rack (30 total circuits) following the same individual-circuit-open → Teams 4 and 5 floor patrol confirm sequence. Team 1 Control provides explicit authorization before each circuit is opened.

39. After all 30 PDU 16 branch circuits are confirmed open and floor patrol checks are clear, Team 2 opens the PDU 16 main breaker and radios confirmation to Team 1 Control.

40. Team 1 Control opens (or directs Team 2 to open) the PDU 16 feeder breaker at MDP-UPS-O/P-3. Team 6 is notified by radio.

> ✅ **VERIFY CHECKPOINT 8:** Confirm PDU 3 load has increased to absorb PDU 16 load. Confirm no equipment is offline in PDU 16-served aisles. Confirm PDU 16 main breaker open. Confirm PDU 16 feeder open at MDP-UPS-O/P-3. Do not authorize Team 6 to begin PDU 16 maintenance until all four conditions are confirmed.

41. Team 3 verifies 0V at PDU 16 main terminals with a calibrated digital multimeter and radios confirmation to Team 1 Control.

42. Team 1 Control applies LOTO at PDU 16 main breaker and PDU 16 feeder breaker at MDP-UPS-O/P-3. Document LOTO application.

43. Team 6, with Blackstone vendor support for PDU 16, performs physical preventive maintenance on PDU 16. Tasks include: dead front panel replacement (estimated 30 minutes), branch circuit maintenance (estimated 15 minutes), internal inspection, cleaning, connection torque verification, and thermal imaging. Team 6 lead and Blackstone technician both sign and record all findings on the maintenance log.

44. Team 6 lead notifies Team 1 Control that PDU 16 maintenance is complete. Team 6 confirms dead front reinstalled, enclosure clear of tools and foreign material, and enclosure door closed.

45. Team 1 Control removes LOTO from PDU 16 main breaker and PDU 16 feeder breaker at MDP-UPS-O/P-3. Document removal.

> ✅ **VERIFY CHECKPOINT 9:** Confirm PDU 16 maintenance complete and signed off by Team 6 and Blackstone. Confirm LOTO removed and documented. Confirm Team 3 monitoring shows no fault conditions at PDU 16 feeder position. Authorize feeder close only after all conditions are confirmed.

46. Team 1 Control closes PDU 16 feeder breaker at MDP-UPS-O/P-3. Team 2 closes PDU 16 main breaker on authorization from Team 1 Control.

47. Team 3 confirms PDU 16 is energized on the monitoring system, voltage is at normal levels, and no alarms are present.

48. Team 2 restores PDU 16 branch circuits one at a time per client rack (30 circuits). Teams 4 and 5 confirm each rack is operational after each circuit is restored.

49. After all 30 PDU 16 branch circuits are restored and all client rack equipment in PDU 16-served aisles is confirmed online, Team 1 Control declares PDU 16 maintenance and restoration complete. Record completion time.

50. Team 1 Control radios all teams confirming PDU 16 is fully restored and authorizes Phase 5 to begin.

> ✅ **VERIFY CHECKPOINT 10:** Confirm all 30 PDU 16 branch circuits restored. Confirm Teams 4 and 5 have cleared all PDU 16-served aisles with no offline equipment found. Confirm monitoring system shows PDU 16 at normal load and no alarms. Record in activity log before proceeding to Phase 5.

PHASE 5 — PDU 17 SHUTDOWN AND MAINTENANCE

51. Team 1 Control authorizes Team 2 to begin PDU 17 branch circuit shutdown sequence. Team 2 proceeds to PDU 17 on the Data Center floor.

52. Team 3 confirms PDU 2 (redundant pair for PDU 17) is live and has confirmed available capacity before any PDU 17 branch circuit is opened.

53. Team 2 opens PDU 17 branch circuits one at a time per client rack (60 total circuits) following the same individual-circuit-open → Teams 4 and 5 floor patrol confirm sequence. Team 1 Control provides explicit authorization before each circuit is opened.

54. After all 60 PDU 17 branch circuits are confirmed open and floor patrol checks are clear, Team 2 opens the PDU 17 main breaker and radios confirmation to Team 1 Control.

55. Team 1 Control opens (or directs Team 2 to open) the PDU 17 feeder breaker at MDP-UPS-O/P-3. Team 6 is notified by radio.

> ✅ **VERIFY CHECKPOINT 11:** Confirm PDU 2 load has increased appropriately. Confirm no equipment is offline in PDU 17-served aisles. Confirm PDU 17 main breaker open. Confirm PDU 17 feeder open at MDP-UPS-O/P-3. Do not authorize Team 6 to begin PDU 17 maintenance until all four conditions are confirmed.

56. Team 3 verifies 0V at PDU 17 main terminals with a calibrated digital multimeter and radios confirmation to Team 1 Control.

57. Team 1 Control applies LOTO at PDU 17 main breaker and PDU 17 feeder breaker at MDP-UPS-O/P-3. Document LOTO application.

58. Team 6 performs physical preventive maintenance on PDU 17. Tasks include: dead front panel replacement (estimated 30 minutes), branch circuit maintenance (estimated 15 minutes), internal inspection, cleaning, connection torque verification, and thermal imaging. Team 6 lead records all findings.

59. Team 6 lead notifies Team 1 Control that PDU 17 maintenance is complete. Confirms dead front reinstalled, enclosure clear, and door closed.

60. Team 1 Control removes LOTO from PDU 17 main breaker and PDU 17 feeder breaker at MDP-UPS-O/P-3. Document removal.

> ✅ **VERIFY CHECKPOINT 12:** Confirm PDU 17 maintenance complete and signed off. Confirm LOTO removed and documented. Confirm monitoring shows no fault conditions at PDU 17 feeder position before authorizing feeder close.

61. Team 1 Control closes PDU 17 feeder breaker at MDP-UPS-O/P-3. Team 2 closes PDU 17 main breaker on authorization.

62. Team 3 confirms PDU 17 energized, voltage normal, no alarms.

63. Team 2 restores PDU 17 branch circuits one at a time per client rack (60 circuits). Teams 4 and 5 confirm each rack is operational after each circuit is restored.

64. After all 60 PDU 17 branch circuits are restored and all client rack equipment in PDU 17-served aisles is confirmed online, Team 1 Control declares PDU 17 maintenance and restoration complete. Record completion time.

65. Team 1 Control radios all teams confirming Phases 2 through 5 (all Data Center floor PDUs) are complete. Teams 4 and 5 complete a final aisle-by-aisle sweep of the entire Data Center 1 floor and confirm all rack equipment is online before proceeding to Phase 6.

> ✅ **VERIFY CHECKPOINT 13:** Confirm PDUs 13, 14, 16, and 17 are all restored and at normal load. Confirm Teams 4 and 5 final floor sweep is complete with no offline equipment identified. Confirm monitoring system shows all four PDUs energized with no alarms. Record final floor clearance confirmation in activity log before proceeding to Phase 6.

PHASE 6 — PDU 13A, PDU 14A, PDU 16A SHUTDOWN AND MAINTENANCE (EE ROOM 3 / TELCO)

66. Team 1 Control authorizes Team 2 to begin shutdown of PDU 13A, PDU 14A, and PDU 16A located in EE Room 3 and the Telco areas of Main Building.

67. Team 3 confirms load monitoring from EE Rooms 1 and 2 is active and that the redundant supply feeds for PDU 13A, PDU 14A, and PDU 16A are live and have confirmed available capacity before any branch circuits are opened.

68. Team 2 turns off PDU 13A branch circuits one at a time. After each circuit is opened, Teams 4 and 5 check the racks in EE Room 3 and EE Room 5 and confirm no equipment has gone offline. Any equipment found offline is immediately reported to the CSG Liaison per the process in Step 10.

69. Team 2 turns off PDU 14A branch circuits one at a time using the same individual circuit open → rack check confirm sequence.

70. Team 2 turns off PDU 16A branch circuits one at a time using the same sequence.

> ✅ **VERIFY CHECKPOINT 14:** Confirm all PDU 13A, 14A, and 16A branch circuits are open. Confirm Teams 4 and 5 have verified no equipment offline in EE Room 3 and EE Room 5. Confirm Team 3 shows load transferred to redundant supplies from EE Rooms 1 and 2. Authorize Team 2 to open main breakers only after all conditions are confirmed.

71. Team 2 opens the PDU 13A main breaker, then the PDU 14A main breaker, then the PDU 16A main breaker. Team 2 radios confirmation to Team 1 Control after each main breaker is opened.

72. Team 1 Control opens (or directs Team 2 to open) the feeder breakers for PDU 13A, PDU 14A, and PDU 16A at MDP-UPS-O/P-3 in EE Room 3. Team 6 is notified by radio that feeders are open.

73. Team 3 verifies 0V at PDU 13A, PDU 14A, and PDU 16A main terminals using a calibrated digital multimeter. Team 3 radios confirmation of 0V on all three PDUs to Team 1 Control before Team 6 is authorized to open any enclosure.

74. Team 1 Control applies LOTO at each of the three PDU main breakers and at their respective feeder breakers at MDP-UPS-O/P-3. Document all LOTO application points in the activity log.

75. Team 6 performs physical preventive maintenance on PDU 13A, PDU 14A, and PDU 16A simultaneously (estimated 45 minutes total for all three). Tasks include: dead front panel replacement, internal inspection, cleaning, connection torque verification, and thermal imaging of all internal connections. Team 6 lead records all findings.

> ✅ **VERIFY CHECKPOINT 15:** Confirm Team 6 has completed maintenance on all three PDUs (13A, 14A, 16A). Confirm dead fronts reinstalled, enclosures clear, and doors closed. Confirm LOTO is ready for removal per Team 1 Control authorization. Do not restore any of the three PDUs until all conditions are confirmed.

76. Team 1 Control removes LOTO from PDU 13A, 14A, and 16A main breakers and their feeder breakers at MDP-UPS-O/P-3. Document removal.

77. Team 1 Control closes the feeder breakers for PDU 13A, PDU 14A, and PDU 16A at MDP-UPS-O/P-3, one at a time, with Team 3 confirming no fault indication between each closure.

78. Team 2 closes the PDU 13A, PDU 14A, and PDU 16A main breakers one at a time on authorization from Team 1 Control.

79. Team 3 confirms all three PDUs are energized on monitoring, voltage is normal, and no alarms are present for any of the three units.

80. Team 2 restores PDU 13A, PDU 14A, and PDU 16A branch circuits one at a time per the same individual circuit close → rack check confirm sequence. Teams 4 and 5 confirm equipment in EE Room 3 and EE Room 5 is operational after each circuit closure.

> ✅ **VERIFY CHECKPOINT 16:** Confirm all PDU 13A, 14A, and 16A branch circuits restored. Confirm Teams 4 and 5 have verified all rack equipment in EE Room 3 and EE Room 5 is online. Confirm monitoring shows all three PDUs at normal load and no alarms. Record Phase 6 completion in activity log before proceeding to Phase 7.

PHASE 7 — UPS SYSTEM 3 AND UPSTREAM DISTRIBUTION SHUTDOWN SEQUENCE

> ⛔ **POINT OF NO RETURN — PHASE 7 INITIATION:** Once the MDP-UPS-O/P-3 main breaker is opened in Step 81, all PDUs served by the MDP-UPS-O/P-3 output (which were previously confirmed de-energized and maintained in Phases 2 through 6) will have no upstream feed path. Confirm all PDU maintenance phases are complete, all client loads are confirmed transferred to redundant feeds, and Team 1 Control has received verbal go-ahead from the DCO facility engineer before proceeding.

81. Team 1 Control (or designated qualified team member at EE Room 3) opens the MDP-UPS-O/P-3 main breaker. This removes output voltage from the entire System 3 output distribution. Team 1 Control radios confirmation of main breaker open status to all teams.

82. Team 3 verifies at the MDP-UPS-O/P-3 output bus that voltage is absent (0V confirmed) before any downstream equipment is handled. Team 3 radios 0V confirmation to Team 1 Control.

83. Team 2 proceeds to UPS Room 3 in Annex Building A-2F. Team 2 presses the **Inverter OFF** button on UPS-3A. Wait for UPS-3A to transition to STS (Static Transfer Switch) mode. Confirm via UPS-3A LCD display that STS mode is active and alarm is present. Do not proceed to UPS-3B until UPS-3A STS mode is confirmed.

84. Team 2 presses the **Inverter OFF** button on UPS-3B. Wait for UPS-3B to transition to STS mode. Confirm via UPS-3B LCD display that STS mode is active and alarm is present. Do not proceed to UPS-3C until UPS-3B STS mode is confirmed.

85. Team 2 presses the **Inverter OFF** button on UPS-3C. Wait for UPS-3C to transition to STS mode. Confirm via UPS-3C LCD display that STS mode is active and alarm is present.

> ✅ **VERIFY CHECKPOINT 17:** Confirm UPS-3A, UPS-3B, and UPS-3C are each individually confirmed in STS mode with LCD active and alarm present on all three units. **CRITICAL NOTE: The UPS output is still live in STS mode. UPS-3A, UPS-3B, and UPS-3C are NOT yet de-energized. Do not treat any UPS as de-energized at this step.** Proceed to Step 86 only after all three STS mode confirmations are received.

86. Team 2 opens the MDP-UPS-I/P-3 main input breaker located in UPS Room 3, Annex Building A-2F. Upon opening this breaker, UPS-3A, UPS-3B, and UPS-3C are now running on battery power only. LCD displays will remain active and alarms will be present. Team 2 radios confirmation to Team 1 Control.

87. Team 3 confirms at the MDP-UPS-I/P-3 that voltage is absent on the output side of the main input breaker. **The UPS systems are still energized from battery at this point. Do not proceed as if the UPS output is de-energized.**

88. Team 2 opens each WAB 3 branch circuit breaker in UPS Room 3 one at a time. After all branch circuits are open, Team 2 opens the WAB 3 main breaker. Team 2 radios confirmation of WAB 3 main breaker open to Team 1 Control.

89. Team 3 verifies 0V at WAB 3 output terminals. Team 3 radios confirmation to Team 1 Control.

90. Team 2 proceeds to Battery Room 3 in Annex Building A-2F. Team 2 opens the battery bank breakers **one at a time**. After each battery bank breaker is opened, Team 2 pauses and confirms no fault condition before opening the next battery bank breaker. Do not open all battery bank breakers simultaneously.

> ✅ **VERIFY CHECKPOINT 18:** Confirm all battery bank breakers in Battery Room 3 are open. Confirm UPS-3A, UPS-3B, and UPS-3C LCD displays have gone dark (no power). Confirm UPS output is de-energized by checking for 0V at UPS output terminals using a calibrated digital multimeter. Only after 0V is confirmed at UPS output terminals are the UPS systems considered fully shut down and safe to work on.

91. Team 3 and the Exquis vendor lead jointly verify with calibrated digital multimeters the voltage on the line side of MDP-UPS-I/P-3, MDP-UPS-O/P-3, WAB 3, and LVMSB-3, and all branch positions. All readings must confirm de-energized (0V) before any maintenance work begins on these panels. Team 3 and Exquis vendor lead both sign the voltage verification checklist.

> ⛔ **POINT OF NO RETURN — TIE-BREAKER OPENING:** Once the TIE-BREAKER at LVMSB-2 is opened in Step 92, LVMSB-3 and all equipment it feeds is fully isolated from the LVMSB-2 upstream source. This action removes the last possible live source feeding System 3 distribution. Confirm Team 3 and Exquis have both signed the voltage verification checklist from Step 91 and that Team 1 Control has received explicit go-ahead from the DCO facility engineer before executing Step 92.

92. Team 2 proceeds to LVMSB Room 2, Annex Building A-2F, and opens the TIE-BREAKER / Bypass Breaker at the LVMSB-2 panel. This isolates LVMSB-3 from LVMSB-2. Team 2 radios confirmation to Team 1 Control.

93. Team 1 Control applies LOTO at the TIE-BREAKER at LVMSB-2, at the MDP-UPS-O/P-3 main breaker, at the MDP-UPS-I/P-3 main input breaker, at the WAB 3 main breaker, and at the LVMSB-3 main position. Document all LOTO application points in the activity log.

94. Team 2 radios all teams to confirm that LVMSB-3, WAB 3, MDP-UPS-O/P-3, and MDP-UPS-I/P-3 are now fully isolated, de-energized, LOTO applied, and are ready for maintenance activity by Team 6.

95. Team 3 and the Exquis vendor lead perform a final voltage verification at LVMSB-3 bus bars, MDP-UPS-O/P-3 bus, MDP-UPS-I/P-3 bus, and WAB 3 bus using calibrated digital multimeters. All readings must confirm 0V before Team 6 is authorized to open any panel for maintenance. Results are recorded and signed on the voltage verification checklist.

> ✅ **VERIFY CHECKPOINT 19:** Confirm TIE-BREAKER at LVMSB-2 is open and LOTO applied. Confirm MDP-UPS-O/P-3 main breaker open and LOTO applied. Confirm MDP-UPS-I/P-3 main input breaker open and LOTO applied. Confirm WAB 3 main breaker open and LOTO applied. Confirm LVMSB-3 LOTO applied. Confirm final voltage verification checklist is signed by both Team 3 lead and Exquis vendor lead showing 0V at all five points. Authorize Team 6 to begin panel maintenance only after all conditions are confirmed and checklist is signed.

PHASE 8 — PARALLEL PANEL MAINTENANCE

Note on parallel execution: The following maintenance tasks are executed in parallel by separate Team 6 sub-teams where Exquis and Blackstone technician staffing permits. Team 1 Control manages all parallel work streams and maintains contact with each sub-team independently. No maintenance task begins without Team 1 Control explicit authorization for that specific equipment item.

96. Team 6 Sub-Team A begins LVMSB-3 maintenance (estimated 80 minutes). Tasks include: dead front panel replacement, internal inspection, bus bar cleaning, connection torque verification per manufacturer specification, contact resistance testing, and thermal imaging of all connections. All findings are recorded by Team 6 Sub-Team A lead.

97. Simultaneously with Step 96, Team 6 Sub-Team B begins MDP-UPS-O/P-3 dead front replacement (estimated 30 minutes). Sub-Team B completes dead front replacement, inspects internal connections, cleans, and torques all connections. Findings recorded.

98. Simultaneously with Steps 96 and 97, Team 6 Sub-Team C begins WAB-3 dead front replacement (estimated 30 minutes). Sub-Team C completes dead front replacement, inspects, cleans, and torques. Findings recorded.

99. Upon completion of MDP-UPS-O/P-3 dead front replacement (Step 97 complete), Team 6 Sub-Team B immediately transitions to MDP-UPS-I/P-3 dead front replacement (estimated 30 minutes, parallel with ongoing LVMSB-3 maintenance in Step 96). Findings recorded.

100. Upon completion of WAB-3 dead front replacement (Step 98 complete), Team 6 Sub-Team C transitions to MDP-UPS-O/P-3 panel maintenance (estimated 35 minutes). Tasks include: detailed internal inspection, connection re-torque, contact cleaning, thermal scan of all positions. Findings recorded.

> ✅ **VERIFY CHECKPOINT 20:** Confirm LVMSB-3 maintenance (Step 96) is ongoing or complete. Confirm MDP-UPS-O/P-3 dead front replacement complete (Step 97). Confirm WAB-3 dead front replacement complete (Step 98). Confirm MDP-UPS-I/P-3 dead front replacement is in progress or complete (Step 99). Confirm MDP-UPS-O/P-3 panel maintenance is in progress or complete (Step 100). Confirm all sub-team leads have radioed status to Team 1 Control. Proceed to Steps 101–105 as each sub-team's prior task completes.

101. Upon completion of MDP-UPS-I/P-3 dead front replacement (Step 99 complete), Team 6 Sub-Team B transitions to WAB-3 panel maintenance (estimated 35 minutes). Tasks include: detailed inspection, connection re-torque, contact cleaning, thermal scan. Findings recorded.

102. Upon completion of MDP-UPS-O/P-3 panel maintenance (Step 100 complete), Team 6 Sub-Team C transitions to LVMSB-3 dead front replacement (estimated 30 minutes, parallel with WAB-3 panel maintenance in Step 101). Findings recorded.

103. Upon completion of WAB-3 panel maintenance (Step 101 complete), Team 6 Sub-Team B transitions to MDP-UPS-I/P-3 panel maintenance (estimated 35 minutes). Tasks include: detailed inspection, connection re-torque, contact cleaning, thermal scan. Findings recorded.

104. Upon completion of LVMSB-3 dead front replacement (Step 102 complete), Team 6 Sub-Team C confirms all LVMSB-3 maintenance (Steps 96 and 102) is complete. Sub-Team C lead radios Team 1 Control confirming LVMSB-3 is ready for restoration.

105. Team 6 lead (overall) collects and reviews all maintenance records from Sub-Teams A, B, and C. Confirms that all panels — LVMSB-3, MDP-UPS-O/P-3, MDP-UPS-I/P-3, and WAB-3 — have completed all maintenance tasks, all dead fronts are reinstalled and secured, all enclosure doors are closed, and no tools or foreign material remain inside any enclosure. Team 6 lead signs the maintenance completion checklist and presents it to Team 1 Control.

> ✅ **VERIFY CHECKPOINT 21:** Confirm Team 6 maintenance completion checklist is signed and received by Team 1 Control. Confirm LVMSB-3, MDP-UPS-O/P-3, MDP-UPS-I/P-3, and WAB-3 are all confirmed: maintenance complete, dead fronts installed, doors closed, clear of foreign material. Confirm MDP-UPS-I/P-3 panel maintenance (Step 103) is complete. Do not authorize any LOTO removal or re-energization until all conditions are confirmed and the signed checklist is in Team 1 Control's possession.

PHASE 9 — UPS SYSTEM 3 AND UPSTREAM DISTRIBUTION RESTART SEQUENCE

106. Team 1 Control authorizes LOTO removal to begin. LOTO is removed in the following order only — do not remove LOTO from downstream equipment before upstream equipment is confirmed ready. Remove

LOTO from the TIE-BREAKER at LVMSB-2 first, then from LVMSB-3, then from MDP-UPS-I/P-3 main breaker, then from WAB-3 main breaker, then from MDP-UPS-O/P-3 main breaker. Document each LOTO removal step in the activity log with time and name of person removing.

107. Team 3 performs a pre-energization inspection of LVMSB-3 externally, confirming the enclosure door is fully closed and latched, no visible damage to the enclosure exterior, and no foreign material at the entry points.

108. Team 2 closes the TIE-BREAKER at LVMSB-2 panel to restore the upstream source feed to LVMSB-3. Team 2 radios confirmation to Team 1 Control immediately upon closure.

109. Team 3 verifies voltage is present at LVMSB-3 bus (400V Line-to-Line confirmed) using a calibrated digital multimeter or panel-mounted metering. Team 3 radios confirmed voltage reading to Team 1 Control. **Do not proceed to close any downstream breaker if voltage is not confirmed at 400V Line-to-Line at LVMSB-3.**

110. Team 2 closes the MDP-UPS-I/P-3 main input breaker in UPS Room 3 on authorization from Team 1 Control. Team 2 radios confirmation.

> ✅ **VERIFY CHECKPOINT 22:** Confirm TIE-BREAKER at LVMSB-2 is closed. Confirm LVMSB-3 voltage verified at 400V Line-to-Line by Team 3. Confirm MDP-UPS-I/P-3 main breaker closed. Confirm monitoring system shows input voltage at MDP-UPS-I/P-3. Do not proceed to WAB 3 restoration or UPS startup until all four conditions are confirmed.

111. Team 2 closes the WAB 3 main breaker in UPS Room 3 on authorization from Team 1 Control. Team 2 radios confirmation.

112. Team 2 closes WAB 3 branch circuit breakers one at a time. After each branch circuit breaker is closed, Team 3 confirms no fault indication before the next branch circuit is closed.

113. Team 3 verifies voltage is present at WAB 3 output and confirms no fault alarms on the WAB 3 panel or monitoring system.

114. Team 2 proceeds to Battery Room 3. Team 2 closes battery bank breakers on UPS System 3 **one at a time**. After each battery bank breaker is closed, pause and confirm no fault indication at the UPS monitoring display before closing the next battery bank breaker.

115. After all battery bank breakers are closed, Team 2 returns to UPS Room 3 and confirms that UPS-3A, UPS-3B, and UPS-3C are showing battery charging status or ready status on their LCD displays.

> ✅ **VERIFY CHECKPOINT 23:** Confirm WAB 3 main breaker closed. Confirm all WAB 3 branch circuits closed. Confirm all battery bank breakers in Battery Room 3 are closed. Confirm UPS-3A, UPS-3B, and UPS-3C LCD displays are active and showing no fault conditions. Do not start UPS inverters until all four conditions are confirmed.

116. Team 2 starts the inverter on UPS-3A by pressing the **Inverter ON** (or equivalent startup command per Exquis procedure for the specific UPS model installed). Wait for UPS-3A to confirm inverter active and output voltage stable. Confirm no alarms on UPS-3A LCD before proceeding.

117. Team 2 starts the inverter on UPS-3B. Wait for UPS-3B to confirm inverter active and output voltage stable. Confirm no alarms on UPS-3B LCD before proceeding.

118. Team 2 starts the inverter on UPS-3C. Wait for UPS-3C to confirm inverter active and output voltage stable. Confirm no alarms on UPS-3C LCD before proceeding.

119. Team 3 verifies at MDP-UPS-O/P-3 that UPS output voltage is present and stable at 400V Line-to-Line at the input side of the MDP-UPS-O/P-3 main breaker before closing.

120. Team 2 closes the MDP-UPS-O/P-3 main breaker in EE Room 3, Annex Building, on authorization from Team 1 Control. Team 2 radios confirmation.

> ✅ **VERIFY CHECKPOINT 24:** Confirm UPS-3A, UPS-3B, and UPS-3C are all confirmed inverter active, output voltage stable, no alarms on LCD. Confirm Team 3 has verified 400V Line-to-Line at MDP-UPS-O/P-3 input side. Confirm MDP-UPS-O/P-3 main breaker is closed. Confirm monitoring system shows UPS System 3 output as energized and at normal parameters. Do not restore PDU feeder breakers until all conditions are confirmed.

121. Team 3 performs a final full voltage sweep of all MDP-UPS-O/P-3 output feeder positions (PDU 13, 14, 16, 17, 13A, 14A, 16A feeder positions) using the panel-mounted metering or calibrated digital multimeter and confirms 400V Line-to-Line is present at each feeder breaker line side before any feeder is closed.

122. Team 1 Control authorizes Team 2 to close PDU 13 feeder breaker at MDP-UPS-O/P-3. Team 2 closes the feeder breaker and radios confirmation. Team 3 confirms voltage present at PDU 13 on monitoring.

123. Team 1 Control authorizes Team 2 to close PDU 14 feeder breaker at MDP-UPS-O/P-3. Team 2 closes and confirms. Team 3 verifies.

124. Team 1 Control authorizes Team 2 to close PDU 16 feeder breaker at MDP-UPS-O/P-3. Team 2 closes and confirms. Team 3 verifies.

125. Team 1 Control authorizes Team 2 to close PDU 17 feeder breaker at MDP-UPS-O/P-3. Team 2 closes and confirms. Team 3 verifies.

> ✅ **VERIFY CHECKPOINT 25:** Confirm feeder breakers for PDU 13, PDU 14, PDU 16, and PDU 17 are all closed at MDP-UPS-O/P-3. Confirm Team 3 monitoring shows 400V at all four PDU positions. Confirm no alarms on any PDU feed position. Authorize PDU main breaker closures only after all conditions are confirmed.

126. Team 2 closes PDU 13 main breaker, then PDU 14 main breaker, then PDU 16 main breaker, then PDU 17 main breaker, one at a time with Team 3 confirming no fault indication between each closure.

127. Team 1 Control authorizes Team 2 to close feeder breakers for PDU 13A, PDU 14A, and PDU 16A at MDP-UPS-O/P-3 one at a time. Team 3 confirms voltage at each feeder position before each closure.

128. Team 2 closes PDU 13A, PDU 14A, and PDU 16A main breakers one at a time on authorization from Team 1 Control. Team 3 confirms each PDU is energized with no alarms after each main breaker is closed.

129. Team 2 restores PDU 13A, PDU 14A, and PDU 16A branch circuits one at a time following the individual circuit close → rack check confirm sequence. Teams 4 and 5 confirm equipment in EE Room 3 and EE Room 5 is operational after each circuit closure.

130. Teams 4 and 5 conduct a complete final walk of all client rack aisles in Data Center 1 floor, EE Room 3, and EE Room 5, and confirm all client equipment is online and operating normally. Team 4 lead and Team 5 lead each radio a final clearance confirmation to Team 1 Control.

> ✅ **VERIFY CHECKPOINT 26:** Confirm PDUs 13, 14, 16, 17, 13A, 14A, and 16A are all energized with normal voltage and no alarms. Confirm Teams 4 and 5 final walk is complete and all client equipment confirmed online. Confirm UPS System 3 is operating normally with inverters active on all three units and no active alarms. Confirm monitoring system shows all in-scope systems at normal operational parameters. This is the final system verification checkpoint. Do not declare procedure complete until all conditions are confirmed.

PHASE 10 — ACTIVITY CLOSE-OUT

131. Team 1 Control collects all maintenance records, voltage verification checklists, LOTO documentation logs, and findings reports from Team 6 lead (Exquis). Confirms all documents are complete, signed, and accounted for.

132. Team 1 Control confirms with the CSG Liaison that no client impact events are unresolved. All client impact events reported during the activity must have a documented resolution or formal open item before close-out.

133. Team 6 lead (Exquis) confirms all tools, test equipment, and materials are removed from all work areas including UPS Room 3, Battery Room 3, LVMSB Room 3, LVMSB Room 2, EE Room 3, and Data Center 1 floor. No tools or foreign material remain in any panel, enclosure, or room where maintenance was performed.

134. Team 1 Control sends a post-activity advisory to the TIM Teams channel and all affected clients confirming the EPM activity at TIM Data Center 1 is complete, all systems are restored to normal operation, and all client equipment is confirmed online.

135. Team 1 Control closes the activity log, records the final completion time, and archives all documentation per facility records retention policy.

EXPECTED OUTCOME: Upon successful completion of this procedure, all in-scope equipment — PDUs 13, 14, 16, 17, 13A, 14A, and 16A; UPS-3A, UPS-3B, and UPS-3C; MDP-UPS-O/P-3; MDP-UPS-I/P-3; WAB-3; and LVMSB-3 — will have completed annual electrical preventive maintenance including dead front replacement, internal inspection, connection torque verification, cleaning, and thermal imaging. All systems will be fully restored to normal energized operation. UPS-3A, UPS-3B, and UPS-3C will be confirmed operating in inverter-active mode with battery banks at normal charge status and no active alarms. All client equipment in Data Center 1 will be confirmed online and operating normally. All four Data Center floor PDUs (13, 14, 16, 17) and all three EE Room PDUs (13A, 14A, 16A) will be confirmed energized at 400V Line-to-Line with all branch circuits restored and serving client racks. A complete set of signed maintenance records, voltage verification checklists, and activity logs will be on file.

IF SOMETHING GOES WRONG